

JAMES RUSE AGRICULTURAL HIGH SCHOOL
TERM 1 ASSESSMENT 1999
YEAR 12 2/3 UNIT

Time allowed: 85 minutes

All questions are to be attempted.
 Approved calculators may be used.
 All questions are of equal value.
 Each question to be handed in separately.

QUESTION 1

- (a) Find the area of the triangle bounded between the x,y axes and the line $3x - 8y = 12$.
- (b) Integrate with respect to x :
- (i) $f(x) = \frac{x}{\sqrt{x}}$
- (ii) $f(x) = \frac{1}{2x-1}$
- (iii) $f(x) = \cos 4x$
- (c) An amount of \$1500 is invested at a rate of 4.5% per annum with the interest compounded monthly. How long will it take for the investment to double its value. (Answer to the nearest month.)

QUESTION 2 (Start a new page)

- (a) The limiting sum of a geometric series is $10\frac{4}{5}$ and the first term is 18.
 Find the fifth term of the series.

- (b) Evaluate:

(i) $\int_0^4 2\pi \, dx$

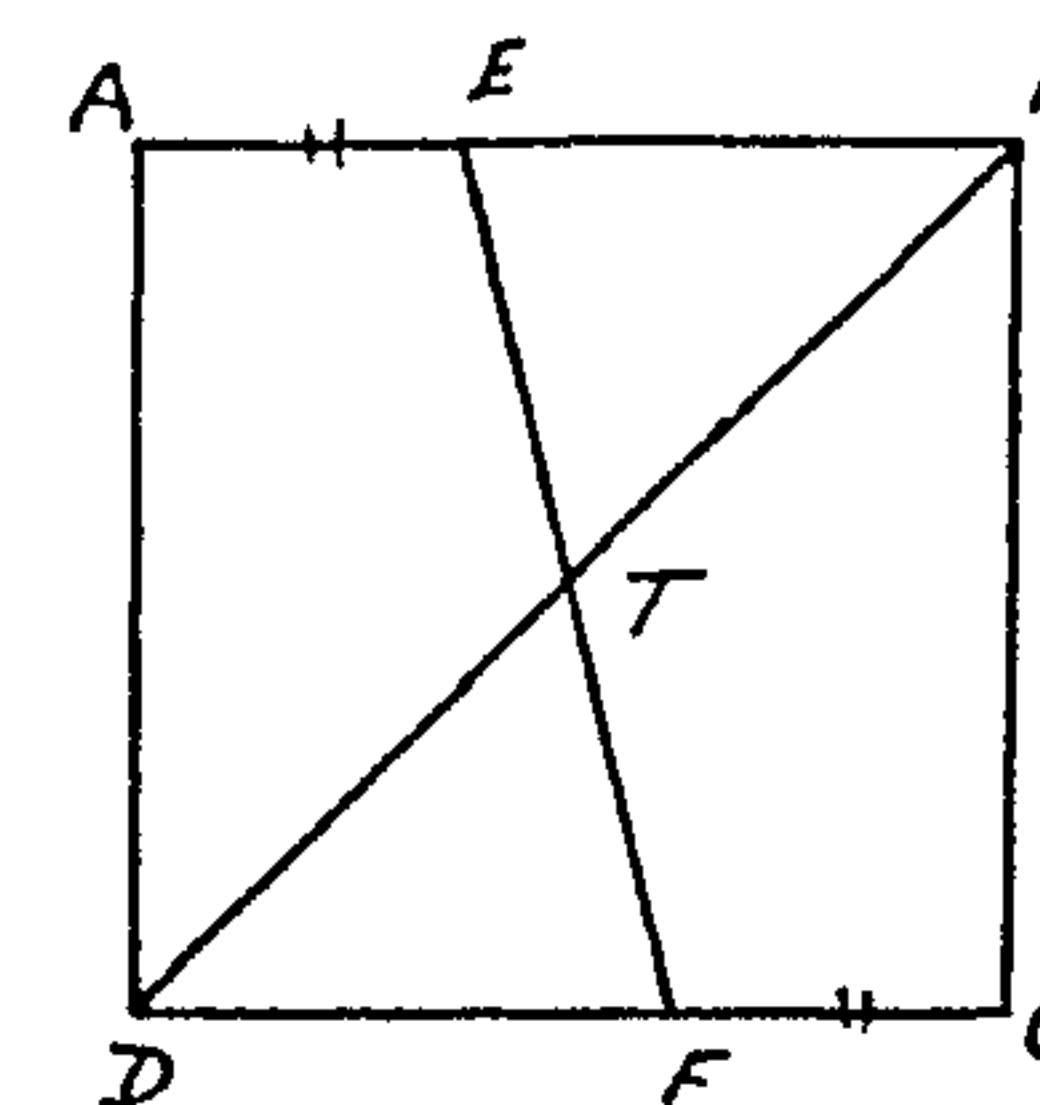
(ii) $\int_1^2 \frac{x-4}{x^2} \, dx$

(iii) $\int_1^4 \frac{1}{\sqrt{x}} e^{\sqrt{x}} \, dx$

QUESTION 3 (Start a new page)

- (a) Find the area between the curve $y = x^2 - 2x$ and the line $y = 2x$.

- (b)



ABCD is a square with $AE = CF$,
 and EF and BD intersect at T.
 Prove that T bisects BD.

**COPY THE DIAGRAM ONTO
 YOUR ANSWER SHEET**

- (c) (i) Draw a neat sketch of the curve with equation $y = \sqrt{4-x}$.
- (ii) Find the area between the curve and the positive x and y axes.

QUESTION 4 (Start a new page)

- (a) (i) A person contributes \$1000 annually in a superannuation fund, commencing at the beginning of 1985. If interest is paid at a rate of 6% per annum, find the value of the investment at the end of year 2010.
- (ii) How many years must the contributor be in the fund in order for the investment to be worth \$100,000.
- (b) Use Simpson's Rule with three function values to approximate the area between the x-axis, the curve $y = e^x \ln(x+1)$, and the lines $x = 0$ and $x = 2$. (Answer correct to two decimal places.)
- (c) The sum of the first n terms of an arithmetic sequence is given by the formula: $S_n = 2n^2 - n$. Find:
- (i) the first term.
- (ii) the sum of the first six terms.
- (iii) a formula for the n th term.

QUESTION 5 (Start a new page)

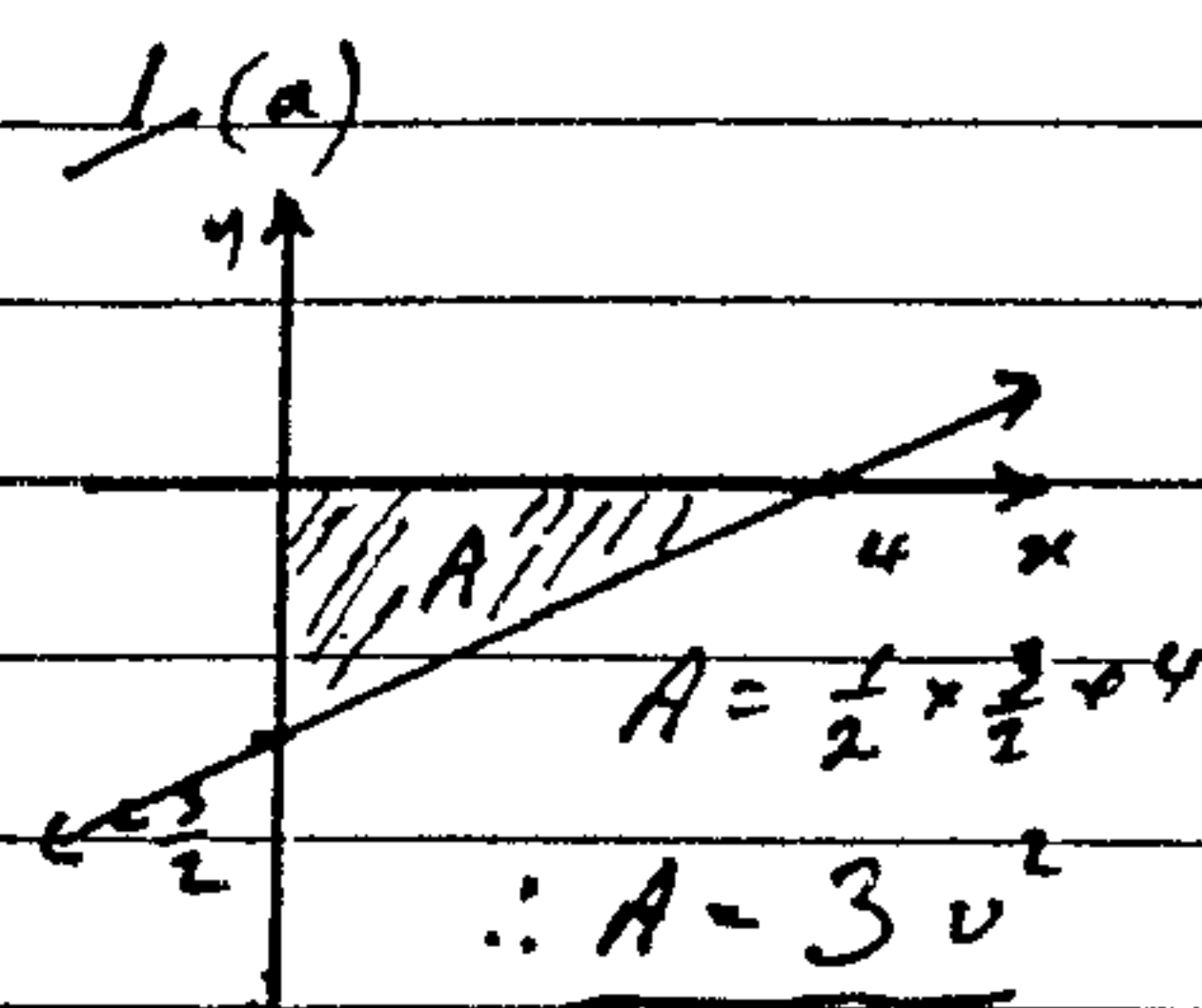
- (a) (i) Draw a neat sketch of the curve with equation $y = \frac{x}{x-2}$, clearly showing all asymptotes and x,y intercepts.
- (ii) Using the graph, or otherwise, find all solutions to $\frac{x}{x-2} > 0$.
- (b) Find the volume generated when the area between the x-axis, the curve $y = \sqrt{\sin 2x}$, and the line $x = \frac{\pi}{8}$ is rotated about the x-axis.
- (c) How many multiples of seven are there between the numbers 500 and 14,495.
- (d) Draw a neat sketch of the curve with equation $y = \sqrt{8+2x-x^2}$.

END of PAPER

TERM 1 ASSESSMENT

YEAR 12 2/3 Unit 1999

SOLUTIONS



(b)(i) $\int \frac{x}{\sqrt{x}} dx = \int \sqrt{x} dx = \frac{2}{3} x\sqrt{x} + C$
 (ii) $\int \frac{dx}{2x-1} = \frac{1}{2} \ln(2x-1) + C$
 (iii) $\int \cos 4x dx = \frac{1}{4} \sin 4x + C$

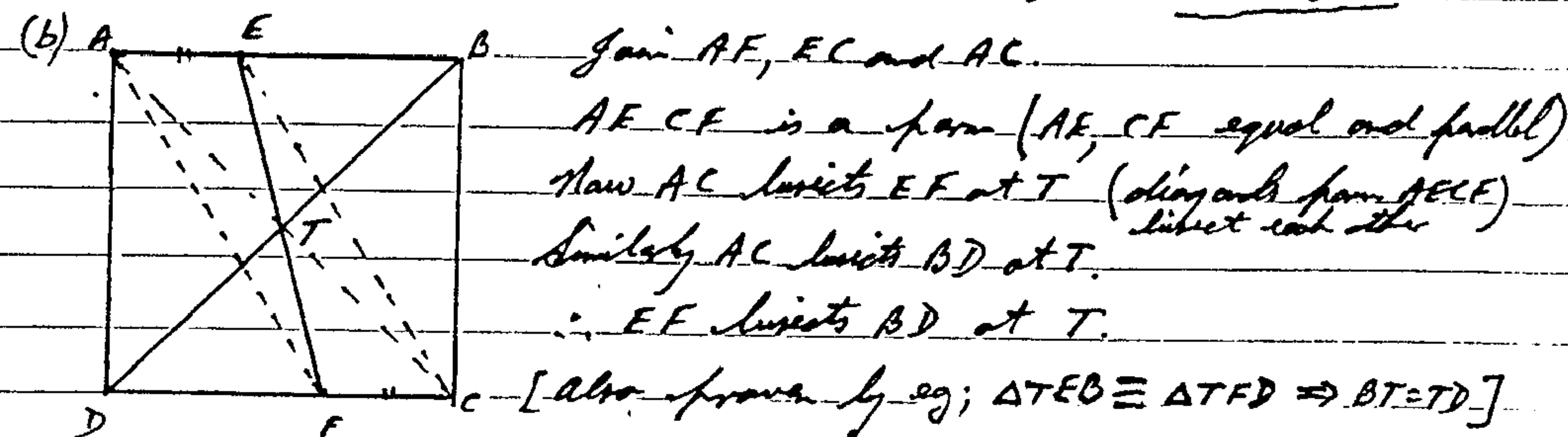
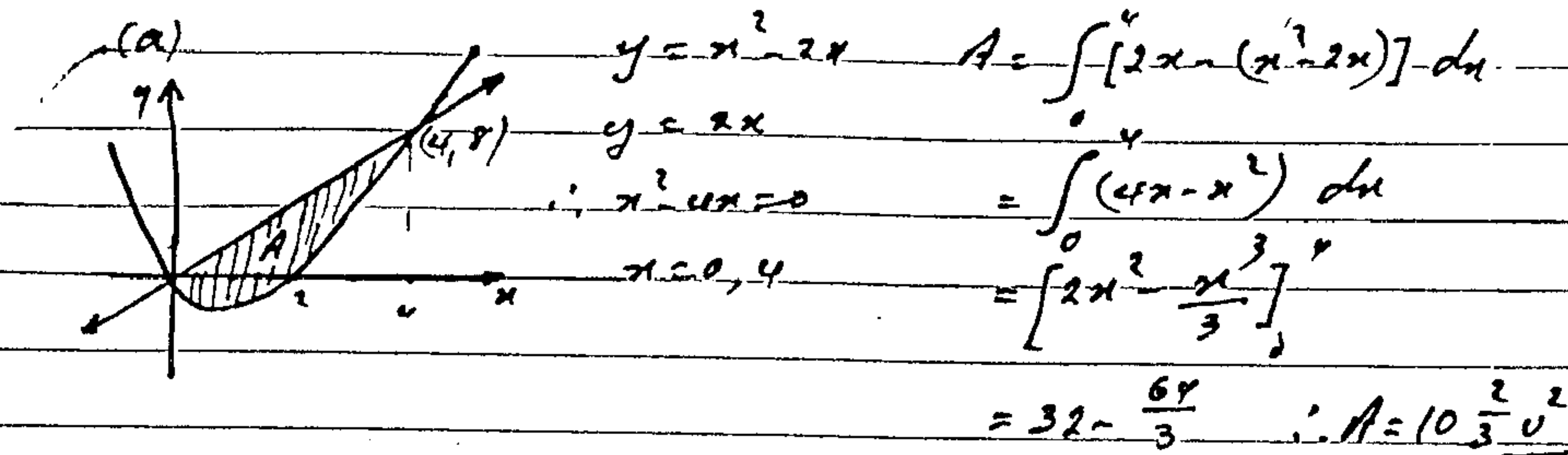
(c) Find t for \$1500 $(1 + \frac{4.5}{100})^t = \3000
 $1.00375^t = 2$
 $t = \frac{\ln 2}{\ln 1.00375}$

$t \approx 186$ months

2. (a) $S_0 = 10 \frac{4}{5}$, $S_1 = \frac{9}{1-r}$
 $a = 18$, $\therefore 10 \frac{4}{5} = \frac{18}{1-r}$
 $\therefore \frac{54}{5} = \frac{18}{1-r}$
 $r = -\frac{2}{3}$

$T_5 = a r^5 = 18 \left(-\frac{2}{3}\right)^5 = 18 \times \frac{16}{81}$
 $\therefore T_5 = \frac{32}{9} (= 3 \frac{5}{9})$

(b)(i) $\int_0^4 2\pi dx = [2\pi x]_0^4 = 8\pi$
 (ii) $\int_1^2 \frac{x-4}{x^2} dx = \int \left(\frac{1}{x} - \frac{4}{x^2}\right) dx = [\ln x + \frac{4}{x}]_1^2 = (\ln 2 + 2) - (\ln 1 + 4) = \ln 2 - 2$
 (iii) $\int_1^4 \frac{1}{\sqrt{x}} e^{\sqrt{x}} dx = 2 \int_1^4 \frac{1}{2\sqrt{x}} e^{\sqrt{x}} dx = 2 [e^{\sqrt{x}}]_1^4 = 2(e - e)$



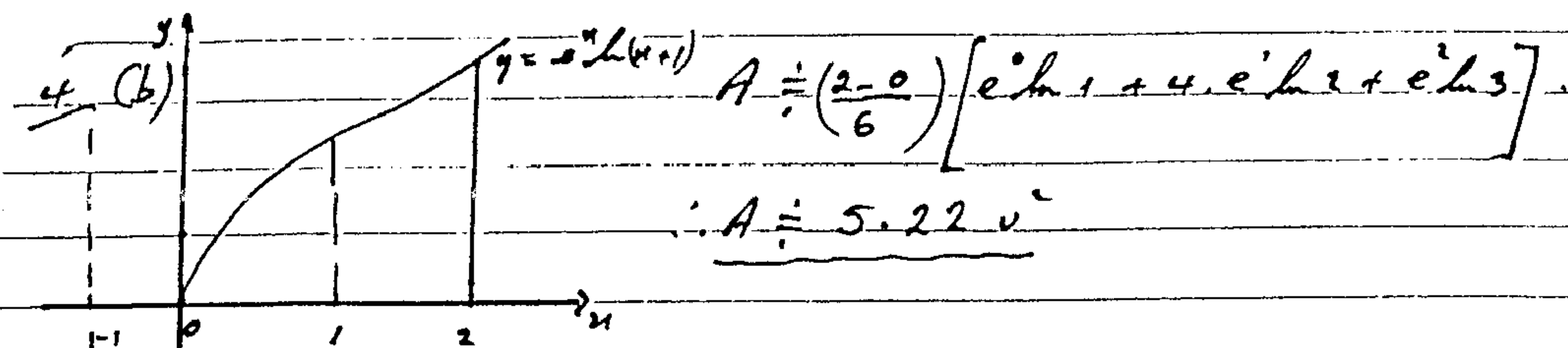
(c)(i) $A = \int_0^4 \sqrt{4-x} dx = [-\frac{2}{3} (4-x)^{\frac{3}{2}}]_0^4 = (-\frac{2}{3} \times 0) - (-\frac{2}{3} \times 4^{\frac{3}{2}}) = 5 \frac{1}{3} u^2$

4. (a)(i) Since beginning of 1985 to end of 2010 is 26 years:
 $I = \$1000 [1.06^{26} + 1.06^{25} + 1.06^{24} + \dots + 1.06]$
 $= \$1000 \times 1.06 \left[\frac{1.06^{26} - 1}{1.06 - 1} \right]$

$\therefore I = \$62,706$

We must find t such that
 $\$1000 [1.06^t + 1.06^{t-1} + \dots + 1.06] = \$100,000$
 $\therefore 1.06 \left[\frac{1.06^t - 1}{1.06 - 1} \right] = 100$
 $1.06^t = \frac{100 \times 0.06}{1.06} + 1$

$\therefore t \approx 32.5$ years



(c) $T_n = S_n - S_{n-1}$

$= (2n^2 - n) - [2(n-1)^2 - (n-1)]$

$= (2n^2 - n) - (2n^2 - 4n + 2 - n + 1)$

$\therefore T_n = 4n - 3$ (iii)

(i) $T_1 = 1$

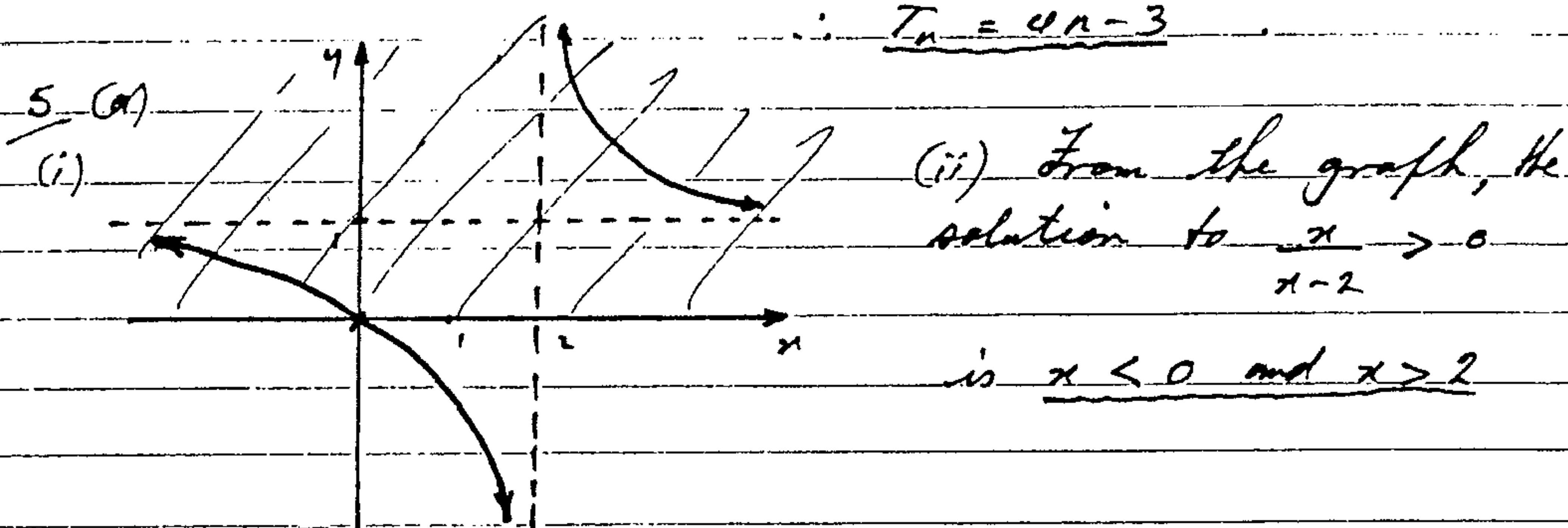
(ii) $S_6 = \frac{6}{2} [2 \cdot 1 + 5 \cdot 4]$

$= 3(2 + 20)$

$\therefore S_6 = 66$

OR (ii) $T_n = 1 + (n-1)4$

$\therefore T_n = 4n - 3$



(b) $V = \pi \int_0^{\frac{\pi}{8}} f(x)^2 dx$

$= \pi \int_0^{\frac{\pi}{8}} \sin^2 x dx$

$= \frac{\pi}{2} [-\cos 2x]_0^{\frac{\pi}{8}}$

$= \frac{\pi}{2} [-\frac{1}{\sqrt{2}} + 1]$

$\therefore V = \frac{\pi}{2\sqrt{2}} (\sqrt{2} - 1) \text{ v}^3$

(c) Between 500 and 14,495 we require terms in $\{504, 511, \dots, 14490\}$

$T_n = a + (n-1)d$

$\therefore 14490 = 504 + (n-1)7$

$\therefore n = 1999$

(d)

$y = \sqrt{8 + 2x - x^2}$

$= \sqrt{9 - (x-1)^2}$

Semi-circle centre at $(1, 0)$ and radius 3.

